

Designing for Storyliving Experiences:

Tinker, A Case Study

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ABSTRACT

Tinker defines a new genre of storytelling called storyliving. Through interactive and embodied immersive experiences, participants can connect with narrative in a more meaningful way. In this live, bespoke virtual reality theatre experience, the participant becomes the main character of the story. Based upon the true story of Director Lou Ward and his relationship with his grandfather who developed Alzheimer's disease, Tinker places the participant in the role of the grandchild so that Ward's story, in turn, becomes their own. Over time, the participant creates new memories alongside their grandfather in his workshop, and their connection to the grandfather and the story grows stronger with time. Tinker is a cornerstone in compelling storyliving experiences; moreover, it is because of how it is designed that participants feel connected to the story.

CCS CONCEPTS

• ; • Interaction design; • Collaborative and social computing; • Human-centered computing;

KEYWORDS

Storyliving, Design, Alzheimer's disease, Virtual Reality

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1 NARRATIVE DESIGN

The core narrative framework behind Tinker is improvisational theatre. This dynamic style of narrative allows the grandfather, portrayed by a professional actor, to generate a more authentic and natural connection with the participant. Utilizing classic improvisational techniques, the actor begins the performance by asking the participant questions about themselves. Those responses are delicately woven throughout the five vignettes in an organic fashion, essentially personalizing the experience and making the participant feel like the story is about them. This not only makes the narrative more compelling for the participant but it makes them feel more emotionally attached to the story. In addition, the unscripted structure allows for free flow dialogue between both characters, establishing a more organic and open platform for communication. This adds an important humanistic element to the virtual performance and allows the participant to feel more comfortable immersing themselves in their role as the grandchild.

2 CHARACTER DESIGN

The grandfather was designed to be an approachable and kind figure. His eyes are intentionally large to support the concept behind the Kindschenschema theory. He is fun-loving and tall, and his avatar is animated like a cartoon to avoid uncanny valley and make him more affable. He is tall with brown hair in the beginning and by the end, he is hunched over, shorter, with pure white hair. The aging process is subtle but vital to convey the passing of time over the course of the vignettes. When Tinker was in development five years ago, the framework for the grandfather avatar was built using motion capture technology on the back end. After the middleware software was bought and deprecated by Apple, the Tinker Studio team pivoted to create their own custom facial gesture controls. Operated exclusively by the actor from their controller, the facial animations range from neutral to confused, to angry, to happy, to sad; the spectrum of emotions encountered by someone struggling with Alzheimer's disease. This animation technology allows the character to make a deeper connection with the participant over the course of the experience by conveying their emotions in a nuanced and effective way.

In addition to the grandfather's customizations, the grandchild also has customized features that allow the participant to feel more connected to the character. In the beginning of the experience, there is a wide range of skin tone colors that the participant can choose from to be reflected in their visible hands, an inclusive design choice that allows the participant to feel more comfortable when embodying their character. The controls also adjust for height so that the participant portrays their accurate height by the end of the experience. In the beginning of the experience, the participant is very small in their crib and has no speaking agency - compared

to the grandfather who is tall and can talk, and shares the world with their small grandchild. In the next scene, the crib is gone and the grandchild is a little taller with full agency. The child continues to grow over the course of the experience, eventually towering over their grandfather. This careful attention to height was necessary to convey the changing perspective and conveyance of particular thematic elements in the narrative and give the illusion of change to the participant to make them feel more immersed in an authentic way.



3 USER INTERACTION DESIGN

The intent of Tinker is creating understanding through play. Over the five vignettes, the grandfather and the grandchild share stories, ideas, and even assets within the virtual workshop! Both characters develop a relationship through collaboration. Whether it is drawing on a whiteboard together, playing with toy cars together, building a Mars Rover together and then watching the rover control camera from the computer, or even taking a Polaroid of moments within the experience, both characters use interaction and collaboration as a form of building connection. The participant has the freedom to play and drive the narrative within the experience, giving them the agency and opportunity to feel empowered as the main character in the story.

4 SOUND DESIGN

The immersive soundscapes in between each vignette are spatialized audio experiences that represent the passing of time and connect the participant to the story in an organic and compelling way. Each soundscape takes place between scenes and is a black void full of stars from the universe, playing on the theme about a joint love for space between the characters. The participant and grandfather can see and talk to one another in this galactical space but it is the sound experience that guides the narrative in this scenario. For example, in the void vignette between Scene 1 and 2, the sound of the grandchild riding the bike, blowing out a birthday cake, making memories with the grandfather over time, signifies the passing of time and setting the stage for the inherent bond that both characters share. Over time, this soundscape becomes a powerful tool to let the participant know that the grandfather's senior moments have evolved to Alzheimer's symptoms, at a time when the grandfather might not recognize or doesn't want to admit the changes going on inside him. Sound is also a powerful tool in a lot of the interactable assets within the environment that help to build connection

between both characters. For example, an answering machine gives the participant insight into how others in the experience are connected to the narrative and in turn, how the participant fits within that ecosystem, and how important they are to the grandfather.

5 USER EXPERIENCE DESIGN

Tinker is an immersive theatre experience for an immersive audience. In addition to the grandfather and the participant, there are about 10 "firefly" characters that have no agency and can't be seen or heard during the vignettes of the performance. They can, however, walk freely throughout the experience as an active and engaged viewer, breaking the boundary present in a typical proscenium theatre. And they can speak in the beginning of the experience when everyone is waiting together in the attic lobby in addition to the end, where everyone returns to the attic lobby for a decompression group session between the grandfather, participant, and viewers. After Tinker's Sundance Film Festival premiere, it was observed that many audience members preferred being a viewer and had just as strong if not stronger of an emotional connection to the story because they could also experience it in an intimate and personal way without being the main character.

6 CONCLUSION

Virtual Reality is often referred to as an "empathy machine". There is tremendous potential to garner understanding from participants through careful user-centric design approaches. Embodied storyliving experiences are an extremely successful modality of connecting participants to the story and allowing them to fully experience a concept, a person, or a story from a new perspective. With Tinker, the goal of the experience was not only to share Lou Ward's story, but allow the story to be experienced by anyone in a very genuine way. So many people's stories are transcendent; it is just a matter of stepping into someone else's shoes to understand their story.